WEEK-10

1.Write a C Program to implement C-SCAN Disk scheduling algorithm.

AIM: To implement C-SCAN Disk Scheduling Algorithm and find Total head movement and Average Head movement.

ALGORITHM:

1.Start.

2.2Read the number of requests, disk size, request queue, and initial head position.

3.Sort the disk requests in ascending order.

4.Display the initial head position.

5.Service all requests greater than or equal to the current head position in ascending order.

6.Move the head to the end of the disk.

7.Move the head from the end to the beginning of the disk without servicing requests during the return.

8.Service the remaining requests in ascending order.

9.Calculate the total head movement.

10.Calculate the average head movement.

11.Display the disk request sequence, total head movement, and average head movement.

12.Stop.

PROBLEM:

VNR VJIET

Name of the Laboratory:

Name of the Experiment:

Experiment No: **Date:**

PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
void sort(int a[], int n){
    int i, j, temp;
    for (i = 0; i < n - 1; i++){
        for (j = 0; j < n - i - 1; j++){
            if (a[j] > a[j + 1]){
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;}}}}
int main(){
    int n, diskSize, head;
    int i, totalSeek = 0;
    int requests[100];
    printf("Enter the number of disk requests: ");
    scanf("%d", &n);
    printf("Enter the disk size: ");
    scanf("%d", &diskSize);
    printf("Enter the disk requests:\n");
    for (i = 0; i < n; i++){
        scanf("%d", &requests[i])}
    printf("Enter the initial head position: ");
    scanf("%d", &head);
    sort(requests, n);
```

```
    printf("\nDisk Request Sequence:\n");
    printf("%d", head);
    for (i = 0; i < n; i++){
        if (requests[i] >= head){
            totalSeek += abs(requests[i] - head);
            head = requests[i];
            printf(" -> %d", head);}}
    if (head != diskSize – 1){
        totalSeek += abs((diskSize - 1) - head);
        head = diskSize - 1;
        printf(" -> %d", head);}
    totalSeek += diskSize - 1;
    head = 0;
    printf(" -> %d", head);
    for (i = 0; i < n; i++){
        if (requests[i] < head){
            totalSeek += abs(requests[i] - head);
            head = requests[i];
            printf(" -> %d", head);}}
    printf("\n\nTotal Head Movement = %d cylinders",    totalSeek);
    printf("\nAverage Head Movement = %.2f cylinders\n", (float)totalSeek / n);
    return 0;
}
```

VNR VJIET

Name of the Laboratory:

Name of the Experiment:

Experiment No: **Date:**

OUTPUT:

TESTCASE1:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc C-scan.c -o C-scan }
 ; if ($?) { .\C-scan }
Enter the number of disk requests: 6
Enter the disk requests:
14
04
62
54
41
34
Enter the initial head position: 50

Disk Request Sequence:
50 -> 54 -> 62 -> 0 -> 4 -> 14 -> 34 -> 41

Total Head Movement = 115 cylinders
Average Head Movement = 19.17 cylinders
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

TESTCASE2:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc C-scan.c -o C-scan }
 ; if ($?) { .\C-scan }
Enter the number of disk requests: 4
Enter the disk requests:
34
56
78
90
Enter the initial head position: 56

Disk Request Sequence:
56 -> 56 -> 78 -> 90 -> 0 -> 34

Total Head Movement = 158 cylinders
Average Head Movement = 39.50 cylinders
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

TESTCASE3:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc C-scan.c -o C-scan }
 ; if ($?) { .\C-scan }
Enter the number of disk requests: 7
Enter the disk requests:
12
34
54
43
65
765
56
Enter the initial head position:
8

Disk Request Sequence:
8 -> 12 -> 34 -> 43 -> 54 -> 56 -> 65 -> 765 -> 0

Total Head Movement = 1522 cylinders
Average Head Movement = 217.43 cylinders
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

2. Write a C Program to implement LOOK Disk scheduling algorithm.

AIM: To implement LOOK Disk Scheduling Algorithm and find Total head movement and Average Head movement.

ALGORITHM:

1.Start.

2.Read the number of disk requests, request queue, initial head position, and disk direction.

3.Sort the disk requests in ascending order.

4.Find the position of the initial head in the sorted request list.

5.If the direction is right:

Service all requests greater than or equal to the head in ascending order.

Then service the remaining requests in descending order.

6.If the direction is left:

Service all requestsess than or equal to the head in descending order.

Then service the remaining requests in ascending order.

7.Calculate the seek time for each movement.

8.Calculate the total and average seek time.

9.Stop.

PROBLEM:

VNR VJIET

Name of the Laboratory: _____

Name of the Experiment: _____

Experiment No: _____ **Date:** _____

PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
void sort(int a[], int n){
    int i, j, temp;
    for (i = 0; i < n - 1; i++){
        for (j = 0; j < n - i - 1; j++){
            if (a[j] > a[j + 1])
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;}}}}
int main(){
    int n, i, head;
    int totalSeek = 0;
    int current;
    int requests[100];
    printf("Enter the number of disk requests: ");
    scanf("%d", &n);
    printf("Enter the disk requests:\n");
    for (i = 0; i < n; i++){
        scanf("%d", &requests[i]);}
    printf("Enter the initial head position: ");
    scanf("%d", &head);
    sort(requests, n);
    current = head;
    printf("\nLOOK Order: %d", current);
```

```
    for (i = 0; i < n; i++){
    if (requests[i] >= head){
            totalSeek += abs(current - requests[i]);
            current = requests[i];
            printf(" -> %d", current);}}
    for (i = n - 1; i >= 0; i--){
        if (requests[i] < head){
            totalSeek += abs(current - requests[i]);
            current = requests[i];
            printf(" -> %d", current);}
    printf("\n\nTotal Seek Time = %d", totalSeek);
    printf("\nAverage Seek Time = %.2f\n", (float)totalSeek / n);
    return 0;
}
```

OUTPUT:

TESTCASE1:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc look.c -o look } ; i
f ($?) { .\look }
Enter the number of disk requests: 9
Enter the disk requests:
56
34
67
36
27
74
5
88
75
Enter the initial head position: 70

LOOK Order: 70 -> 74 -> 75 -> 88 -> 67 -> 56 -> 36 -> 34 -> 27 -> 5

Total Seek Time = 101
Average Seek Time = 11.22
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

TESTCASE2:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc look.c -o look } ;
f ($?) { .\look }
Enter the number of disk requests: 7
Enter the disk requests:
34
87
54
34
76
32
59
Enter the initial head position: 78

LOOK Order: 78 -> 87 -> 76 -> 59 -> 54 -> 34 -> 34 -> 32

Total Seek Time = 64
Average Seek Time = 9.14
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

TESTCASE3:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc look.c -o look } ; i
f ($?) { .\look }
Enter the number of disk requests: 6
Enter the disk requests:
89
25
65
06
36
26
Enter the initial head position: 76

LOOK Order: 76 -> 89 -> 65 -> 36 -> 26 -> 25 -> 6

Total Seek Time = 96
Average Seek Time = 16.00
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

3.Write a C Prpgram to implement C-LOOK Disk Scheduling Algorithm.

AIM: To implement C-LOOK Disk Scheduling Algorithm and find Total head movement and Average Head movement.

ALGORITHM:

1.Start.

2.Read the number of disk requests, request queue, and initial head position.

3.Sort the disk requests in ascending order.

4.Find the requests greater than or equal to the initial head.

5.Service these requests in ascending order.

6.After reaching the highest request, jump to the smallest request.

7.Service the remaining requests in ascending order.

8.Calculate the seek time for each movement, including the circular jump.

9.Calculate the total and average seek time.

10.Stop.

PROBLEM:

VNR VJIET

Name of the Laboratory:

Name of the Experiment:

Experiment No: **Date:**

PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
void sort(int a[], int n){
    int i, j, temp;
    for (i = 0; i < n - 1; i++){
        for (j = 0; j < n - i - 1; j++){
            if (a[j] > a[j + 1]){
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;}}}}
int main(){
    int n, i, head;
    int totalSeek = 0;
    int current;
    int requests[100];
    printf("Enter the number of disk requests: ");
    scanf("%d", &n);
    printf("Enter the disk requests:\n");
    for (i = 0; i < n; i++){
        scanf("%d", &requests[i]);}
    printf("Enter the initial head position: ");
    scanf("%d", &head);
    sort(requests, n);
    current = head;
    printf("\nC-LOOK Order: %d", current);
```

```
    for (i = 0; i < n; i++){
        if (requests[i] >= head){
            totalSeek += abs(current - requests[i]);
            current = requests[i];
            printf(" -> %d", current);}}
    for (i = 0; i < n; i++) {
        if (requests[i] < head) {
            totalSeek += abs(current - requests[i]);
            current = requests[i];
            printf(" -> %d", current);}}
    printf("\n\nTotal Seek Time = %d", totalSeek);
    printf("\nAverage Seek Time = %.2f\n",
          (float)totalSeek / n);
    return 0;
}
```

OUTPUT:

TESTCASE1:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc c-look.c -o c-look }
 ; if ($?) { .\c-look }
Enter the number of disk requests: 6
Enter the disk requests:
22
3
33
44
55
66
Enter the initial head position: 14

C-LOOK Order: 14 -> 22 -> 33 -> 44 -> 55 -> 66 -> 3

Total Seek Time = 115
Average Seek Time = 19.17
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

TESTCASE2:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc c-look.c -o c-look }
 ; if ($?) { .\c-look }
Enter the number of disk requests: 6
Enter the disk requests:
99
88
77
66
55
44
Enter the initial head position: 50

C-LOOK Order: 50 -> 55 -> 66 -> 77 -> 88 -> 99 -> 44

Total Seek Time = 104
Average Seek Time = 17.33
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```

TESTCASE3:

```
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS> cd "c:\Users\harsh\
OneDrive\Documents\OS_25071A6231\OS PROGRAMS\" ; if ($?) { gcc c-look.c -o c-look }
 ; if ($?) { .\c-look }
Enter the number of disk requests: 5
Enter the disk requests:
78
34
65
78
54
Enter the initial head position: 12

C-LOOK Order: 12 -> 34 -> 54 -> 65 -> 78 -> 78

Total Seek Time = 66
Average Seek Time = 13.20
PS C:\Users\harsh\OneDrive\Documents\OS_25071A6231\OS PROGRAMS>
```